Structural Non-Isolability in Knowledge-Based Robot Fault Diagnosis: Tie-Break Intervals and Grouped Validation

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Abstract: Coupling a learned anomaly detector to a line-specific FMEA knowledge base gives a robot cell a ranked list of candidate causes. We show that on the public voraus-AD benchmark this ranking is structurally unable to isolate: the knowledge base's fault signature matrix has repeated columns, and 514 of 755 anomalies have an exact score tie at rank one. Three measurements follow. First, a prevalence prior that reads no evidence outscores the deployed retrieval on every ranking metric, so the customary lift over uniform chance is an artefact of the reference. Second, permuting only the order in which failure modes are declared moves the reported top-1 across an interval 0.082 wide, more than twice the spread the validation protocol itself contributes, while the cluster assignment is invariant at 0.342 under every ordering we tried. Third, the protocol decides deployment: the benchmark's `setting` values are repeated instantiations of single physical faults, and under an ungrouped split the identical pipeline certifies a collision repair at 0.830 that leave-setting-out validation rejects. We therefore report the non-isolable groups rather than breaking them, and search a grammar of physically named representation dimensions for the one that separates each group, the representation-space analogue of sensor placement for isolability. One of four groups is repaired with held-out evidence (0.840 against 0.562); the others return *inconclusive*, *undetermined* and *hardware gap*. A flat classifier over the same features outscores the whole pipeline on point accuracy, which we report with the label budget at which it does so.

Keywords: Fault isolation and root cause analysis, Structural analysis methods, Diagnosability analysis, Knowledge-based diagnosis, Anomaly detection, Robotics.

1. Introduction

A learned anomaly detector on robot telemetry reports that a cycle is abnormal. It does not report why. The obvious way to close that gap is a line-specific FMEA knowledge base: each failure mode declares the joints it affects and the signal families in which it manifests, a retrieval function scores the modes against the detector's per-signal evidence, and the maintenance technician receives a ranked list.

Where no analytical model of the cell exists, data-driven detection is now the routine substitute for model-based residual generation, and it brings with it the habit of reporting a single accuracy figure against uniform chance. This paper is a measurement of what such a ranked list can and cannot decide on the public voraus-AD benchmark (Brockmann et al., 2024), summarised in Fig. 1. The result is negative in a specific and useful way. The knowledge base's declarations form a fault signature matrix in the sense of Blanke et al. (2016), and that matrix has repeated columns: nine of fourteen failure modes share a signature with at least one other. Repeated columns are the classical statement of non-isolability, and they do not become isolable because the scoring function is continuous. What the scoring function does instead is produce an exact tie, in 514 of 755 anomalies, which some arbitrary rule then breaks.

Two things follow that we had not expected and that we believe generalise beyond this benchmark.

The first concerns references. Diagnostic accuracy on a skewed benchmark is routinely compared with uniform chance. On voraus-AD the correct reference is a prevalence prior (rank the modes by how often they occurred in the fit half, then emit that same ranking for every cycle), and that prior outscores the deployed retrieval on every ranking metric. A reported lift of 2.37× over uniform chance is 0.91× against the prior.

The second concerns validation. voraus-AD's `setting` field indexes physical fault instantiations, each repeated over several cycles. Splitting cycles at random puts the same instantiation on both sides of the split. We show that this does not merely inflate a number: it reverses a deployment decision. The same pipeline that adopts a repair to the distal-collision cluster under a random split rejects it under leave-setting-out validation, and no permuted-label control catches the difference. Leakage of this kind is a documented failure mode across machine-learning-based science (Kapoor and Narayanan, 2023), and time-series

anomaly-detection benchmarks in particular have been shown to overstate progress (Wu and Keogh, 2023).

Contributions. (i) A structural indistinguishability analysis of a learned diagnostic representation, computed as a fault signature matrix and reported as non-isolable groups rather than broken by a tie-break rule. (ii) Two references this benchmark's literature has not applied, a prevalence prior and a tie-break interval, against which the deployed ranking trails the first and is swamped by the second. (iii) A repair that searches physically named representation dimensions for the one separating a group, as the representation-space analogue of sensor placement, issued as a certificate with held-out evidence, controls and a cost class. (iv) Evidence that grouped validation changes deployment decisions, not only reported accuracy. (v) A control that outscores our method, with the label budget above which it does so.

2. Fault signatures and isolability

Structural diagnosability analysis determines which faults a diagnostic system can separate before any data is seen, from the incidence structure of the residuals against the fault modes (Blanke et al., 2016; Krysander et al., 2008). The standard object is the fault signature matrix, whose columns are fault modes and whose rows are residuals or tests: two modes are non-isolable exactly when their columns coincide. The technique has been applied to public process benchmarks in this same diagnostic spirit (Düstegör et al., 2006) and remains active (Travé-Massuyès and Wotawa, 2024); quantitative extensions replace the binary verdict with a graded measure of distinguishability under noise (Jung et al., 2015).

We borrow the output form. In the FDI tradition a diagnostic system emits a decision; in the DX tradition a diagnosis is a *set* of candidates consistent with the observations (Reiter, 1987; de Kleer and Williams, 1987), and the formal correspondence between the two is by now well understood (Cordier et al., 2004; Travé-Massuyès, 2014). Our observation is that a system built on the FDI side of that correspondence, with a learned detector in place of an analytical model, produces sets whether or not it admits to it (the sets are the tied candidates), and that scoring it as though it emitted decisions measures the arbitrary rule that collapses them.

When a group of modes is found non-isolable, the classical remedy is sensor placement: determine which additional measurement would make the group isolable, and at what cost (Krysander and Frisk, 2008; Frisk et al., 2009). Our setting inherits the question and changes the search space. There is no analytical model of the cell, and the rows of our signature matrix are not residuals of an observer but declarations in an FMEA knowledge base: affected joints and manifesting signal families. The remedy we can afford is correspondingly different: not which sensor to add, but which dimension of the telemetry the deployed representation currently discards. The cost hierarchy of Section 6 plays the role sensor cost plays in the classical formulation.

3. System and data

Benchmark. voraus-AD (Brockmann et al., 2024) contains machine data from a six-axis robot on a pick-and-place task, with 12 anomaly types physically induced. We use the 100 Hz release: 2122 cycles, of which 755 are anomalous and 1367 normal. The official training split is the 948 normal runs recorded under `setting` $= 72$; evaluation uses the disjoint `setting` $\neq 72$ split (419 held-out normal cycles). The anomalous cycles fall under 72 distinct `setting` values, each identifying one physical fault instantiation repeated for 4–20 cycles, median 10.

Detector. The benchmark's own baseline is a normalizing flow over robot telemetry (Brockmann et al., 2024), one instance of the density-estimation family surveyed by Ruff et al. (2021). We use a light CPU flow over 46 measured channels, reaching AUROC 0.854 and, with the threshold set at the 95th percentile of a calibration half of the held-out normal cycles, TPR 0.601 and, on the disjoint evaluation half, FPR 0.071; the 0.05 the calibration fixes holds in sample only. This is below the 0.936 reported for the original model; the gap is architectural rather than informational, and detection accuracy is not this paper's claim. Every result below is conditioned on a cycle already flagged, and the detector supplies a verdict plus a per-signal standardised residual.

Knowledge base and retrieval. The knowledge base is a conventional FMEA encoded as a graph; recent work automates the construction of such graphs for robotic diagnosis (Liao et al., 2025), but ours is small and hand-authored, the realistic case for a single commissioned line. It declares 14 failure modes, including the 12 voraus-AD anomaly types, each with affected joints, manifesting signal families, a cycle-phase hint and a recommended check. Retrieval scores a mode m against a cycle record r as a weighted sum of Jaccard overlap of manifesting families gated by affected joints, an inverse-document-frequency-weighted signal evidence term, phase agreement and manifest coverage. Weights are fixed a priori; re-tuning them on a stratified 50/50 fit half over five seeds converges to the same weights on 5/5 seeds and gives held-out top-1 0.158 ± 0.006 (95% interval over seeds), so what follows is not tuning on test.

4. Structural indistinguishability

A mode's *structural signature* is the pair (declared affected-joint scope, declared manifesting signal families). Two modes with the same signature are interchangeable under the deployed representation however the retrieval weights are set, because every term in the score is a function of that pair and of the record. Fig. 2 shows the matrix.

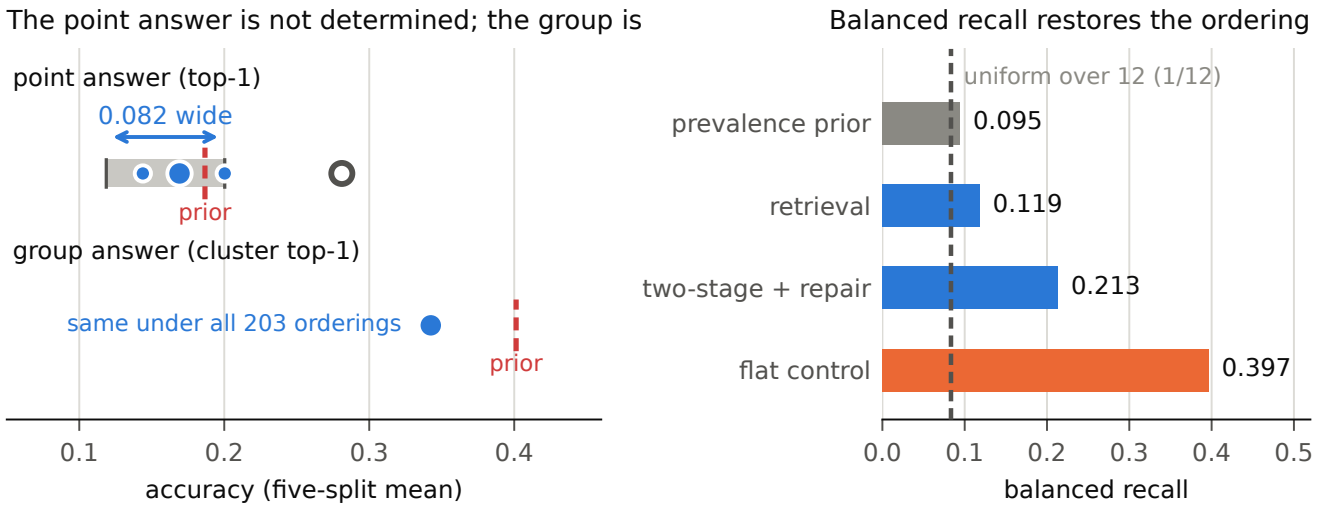


Figure 1: The measurement this paper reports. *Left:* the point answer depends on an arbitrary rule. The band spans 200 random orderings of the knowledge base's mode declarations, the filled markers the reverse, declaration and alphabetical orderings, the open marker an oracle bounding what any tie-break could achieve. Every mark and dashed prior is a five-split mean over the splits of Table 1. The group assignment below does not move at all. *Right:* against balanced recall (the mean of twelve per-category recalls, which no constant predictor can inflate) the ordering is restored and the repair carries information the prior does not.

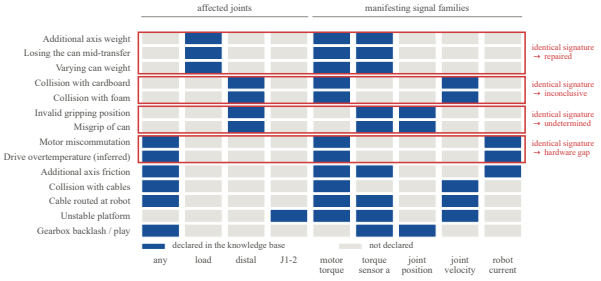


Figure 2: Structural signatures of the 14 failure modes. The four bracketed groups share a signature and are therefore non-isolable under this representation. Gearbox backlash declares the same two signal families as the gripping pair but a different joint scope, so it is not a twin — the signature is the pair, not the families alone.

Nine of the fourteen modes fall into four groups: three load-torque modes (axis weight, lost payload, variable weight), two distal collisions (carton, foam), two gripping faults (invalid grip, mis-grip) and a commutation/over-temperature pair. This is a claim about the knowledge base, not about physics: two modes with the same signature may well differ in the raw telemetry, and the repair of Section 6 exploits that difference.

In 514 of 755 anomalies (68.1%) the rank-1 candidate lies in one of these groups, and every one of those is an exact score tie: 453 of size two, 61 of size three. None arises from score quantisation. We route each cycle by a cluster-margin test applied before a cluster-size test, the reject option of Chow (1970): 122 cycles (16.2%) fall below the margin and are declared *cluster-undetermined*, 134 (17.7%) have a rank-1 mode alone in its group and are *decisive*, and 499 (66.1%) are *human-resolvable*. Cluster accuracy by route is 0.180, 0.187 and 0.421 in that order, reproducing the full-set cluster accuracy of 0.340 exactly when weighted by route size; Table 1 gives the corresponding five-split mean, 0.342 ± 0.025. The two are the same quantity over different populations, and we distinguish them wherever both appear. Note that the decisive route is no more accurate than the undetermined one: a rank-1 mode alone in its group is not thereby more often right.

5. What the reported accuracy is not

5.1. The prevalence prior

The reference customarily reported for a ranked candidate list is uniform chance, 0.071 over 14 modes, against which the deployed retrieval shows a lift of 2.37×. Uniform chance is the wrong reference for a skewed benchmark, in which category 1 contributes 156 cycles and category 8 contributes 10. The reference such a benchmark demands is a *prevalence prior*: rank the modes by fit-half frequency and emit that ranking for every cycle. It reads no evidence and cannot distinguish one cycle from another.

The prior beats the deployed retrieval on every ranking metric (Table 1) and also on cluster top-1, so retreating to the cluster level does not escape it. Against that reference the retrieval's lift is 0.91×.

The one reading a prior cannot game is *balanced recall*, and there the ordering is restored and monotone (Table 1), so the retrieval does carry per-cycle information. It carries very little: its recall is zero on seven of

Table 1: Deployed retrieval against two trivial baselines, over five leave-setting-out splits. The prior wins on all four ranking metrics (top-1, top-3, top-5 and MRR; two shown for space) and on cluster top-1, and loses only on balanced recall, the mean of the per-category recalls. A constant predictor is bounded there by $1/K$, with K the number of categories in the test half: three categories have a single fault instantiation each, so under leave-setting-out they fall wholly on one side of the split and K is 10 or 11, not 12. That bound averages 0.095 over the five splits, which is what the prior attains; a uniform predictor over the twelve categories scores $1/12 = 0.083$ for any K.

	top-1	top-5	cluster	balanced
uniform chance	0.071	0.357	0.150	0.083
prevalence prior	**0.187**	**0.705**	**0.401**	0.095
retrieval (deployed)	0.169	0.651	0.342	0.119
two-stage (Sec. 6)	0.309	—	0.342	0.213
flat control (Sec. 8)	0.525	—	—	0.397

the twelve categories, and on the two largest it is worse than the constant (0.238 against 0.400; 0.277 against 0.600). For that reason the comparisons between methods below rest on balanced recall; Section 5.2 stays with top-1 because that is the number this literature reports.

5.2. The tie-break interval

Which member of a tied set is reported is not determined by the ranking. In our implementation it is determined by the order the modes happen to be declared in the knowledge base. Varying only that order on the splits of Table 1 (Fig. 1, left): the deployed declaration order gives 0.169, alphabetical 0.200, reverse declaration 0.144, and 200 random orderings give 0.159 (s.d. 0.027), range $[0.119, 0.200]$. An oracle that always picks the correct member of the tied set reaches 0.281, which bounds what any tie-breaking rule could achieve.

The interval is 0.082 wide. The same five-split mean has a 95% interval of ± 0.018 across splits (Section 6), so the arbitrary rule contributes more than twice the spread the validation protocol does, and the third decimal of a reported top-1 on this benchmark is not a property of the method. Both spreads, 0.036 across splits and 0.082 across orderings, exceed the 0.018 by which the deployed retrieval trails the prevalence prior in Table 1, and the interval contains the prior: alphabetical order alone would put the retrieval ahead of it.

This is the paper's thesis in numerical form. It has a positive counterpart: the *cluster* assignment is tie-break invariant, at 0.342 under all 203 orderings, because the ties lie inside groups and never across them. What the representation determines is the group, not the point answer.

6. Sensor placement over representation dimensions

For each non-isolable group we ask which named quantity, if the representation carried it, would separate the members.

Grammar. Candidate dimensions carry physical names: sign of deviation, within-cycle step, a carry-versus-empty payload contrast, dispersion, and channels in the data but never read by the attribution. Each carries a cost class: *free* if already in the evidence record, *pipeline* if in the data but unread, *hardware* if not measured in the cell at all. The last is where the classical answer returns: buy a sensor.

Certificate. A repair is not applied but issued as a certificate stating the dimensions added, their cost class, within-group accuracy on unseen fault instantiations, and the three controls of Table 2. It stays inactive until an engineer authorises it, and is credible only if the repaired figure exceeds all three.

Table 2 gives the four verdicts, and we distinguish the three non-positive ones because they are not the same claim. **Adopted**: the load-torque group is separated at 0.840 on held-out instantiations against 0.562 for the deployed representation, a 0.504 majority floor and 0.431 under permuted labels, on all five splits; its cost class is *pipeline*, as unread channels enter on two. **Inconclusive**: distal collision has too few fault instantiations for the grouped inner search to run, so its 0.776 comes from the pre-registered free-cost dimension set rather than from a search. It exceeds the 0.718 majority rate by less than its own sampling error: with $n_{\text{test}} = 57$ the binomial 95% half-width is ≈ 0.108. **Undetermined**: gripping has one instantiation per member, so no grouped split exists to evaluate anything on. The distinction from the row above is between a grouped *split* and a grouped *search*; collision admits a split, so a pre-registered dimension set can at least be evaluated on it. This is a statement about the evidence, not the world, and Section 8 measures how many instantiations would be needed. **Hardware gap**: no temperature channel exists in the cell, so no dimension of the grammar could separate commutation faults from drive over-temperature. Of the four verdicts this is the only claim about the plant itself, and it is the classical sensor-placement answer.

End to end, the two-stage system moves point top-1 from 0.169 ± 0.018 to 0.302 ± 0.047 against a stage-1 cluster ceiling of 0.342 ± 0.025, and balanced recall from 0.119 to 0.213; the fixed shipping configuration reported in Table 1 reaches 0.309 ± 0.031 and 0.869 within the load-torque group. Both are the same metric on the same held-out halves ($n_{\text{test}} = 152$): the searched certificate re-selects dimensions per split and pays the generalisation cost of that selection, the fixed configuration none. We headline the certificate because it must include the cost of finding the repair. Intervals are $t_{0.975,4} = 2.776$ standard errors of the split mean; a

Table 2: Repair certificates. "Grouped" is leave-setting-out validation and is the figure we report; "Ungrouped" is the identical pipeline under a random split over cycles. "Deployed" is the current representation on the same held-out half and "Permuted" the same search with shuffled labels, both grouped. Only the load-torque group clears all three controls, and for the collision group the choice of protocol changes the verdict itself, not just the number. †Collision figures come from the pre-registered free-cost dimension set: its instantiation count is too low for the grouped inner search to run, so this row is not the same procedure as the one above it.

Group	Ungrouped	Grouped	Deployed	Majority	Permuted	Verdict
load torque (3 modes)	0.933	0.840	0.562	0.504	0.431	**adopted**, both protocols, 5/5 splits
distal collision (2 modes)†	0.830	0.776	0.709	0.718	0.608	**inconclusive** grouped; adopted ungrouped
gripping (2 modes)		one instantiation per member				undetermined
commutation/temp (2 modes)		no temperature channel exists				hardware gap

normal approximation would understate them by about 30%.

7. The validation protocol decides deployment

Each `setting` value is one physical instantiation of a fault, so a random split over cycles places the same instantiation on both sides and a model can recognise the session rather than the fault. We hold out whole `setting` values and require that a certificate's evidence come from instantiations the search never saw. The two protocols separate the arms as their label dependence predicts: the flat control falls from 0.800 to 0.525, while retrieval improves slightly (0.162 to 0.169) because it reads no labels and cannot benefit from leakage.

Table 2's first two columns carry the result that matters. The load-torque certificate survives the change of protocol. The distal-collision group does not: under a random split it is certified at 0.830 and adopted, and under leave-setting-out it reads 0.776 and is rejected. Nothing in the ungrouped run looks wrong: the permuted-label control passes, the margin over the deployed representation is comfortable, and the verdict is stable across seeds. What separates a recommendation to change what a plant measures from a session-recognition artefact is the protocol, not the method. We therefore state it as a requirement: any learned comparison on this benchmark should group by `setting`.

8. Controls, and what would settle the open verdicts

8.1. The flat classifier

A flat 12-way classifier over all grammar features reaches 0.525 under the same grouped protocol, well above our 0.309, and 0.397 balanced recall against our 0.213. We tested the label-efficiency defence one would reach for and it did not survive: with both systems given the same labels and scored end to end on the same twelve-way task, the control leads from $k = 8$ labels upward and the gap widens (0.226 against 0.196 at $k = 8$; 0.526 against 0.309 at $k = 320$). Below $k = 8$ neither arm separates from the other, or from a rule that predicts the majority category of its own k labels, so the small-budget region is not a regime either method wins. The full curve is in the repository.

Wherever labels can be collected, a flat classifier over the same features is the stronger choice for point accuracy; the commissioning regime we had hoped to claim for ourselves is one in which no method here is measurably better than any other.

8.2. Where the ranking's output goes

The load-torque group is the true answer on 41.1% of cycles but is ranked first on 66.0%, an over-prediction that consumes most of the ranking's output. The distal-collision and gripping groups are the true answer on 12.5% and 3.0% and are ranked first on *none*, so their certificates are computed over cycles selected by true label and state whether a repair would hold if the group were reached. They do not describe deployed behaviour. This sharpens the control result: the flat classifier recovers foam collision at 0.506, a category on which the retrieval and the prior both score zero. The evidence is in the same envelope the grammar features are computed from; the attribution and retrieval layer never reaches it.

8.3. How many instantiations would settle it?

An *inconclusive* verdict is a confession unless one can say what would resolve it. On real data the instantiation count is fixed at whatever the benchmark supplies, so we make it a knob and sweep it on injected faults.

No URDF of the cell exists, so payload torque is measured: carry-phase samples are matched to their nearest empty-phase neighbours in six-joint space and differenced, and the calibration is in the repository. Because the fault is injected along a dimension of the same grammar the search examines, *nothing here may be used to claim that repair works.* We declare that circularity at the source and claim only a property of the protocol.

On the gripping pair, $N \leq 3$ instantiations leave the grouped inner search unable to run; at $N = 4$–6 it runs but the verdict flips between draws; from $N \geq 8$ it

is stable. Note, though, what it stabilises *on*: reject. The deployed representation already comes within 0.02 of the repaired ≈ 0.99, so whether the same threshold governs an *adopt* verdict is untested. The collision pair reaches the run threshold at the same $N \geq 4$ but is still flipping at $N = 16$. voraus-AD supplies $N = 1$ per gripping mode and $N = 3$ for carton collision, both below the $N \geq 4$ the search needs. *The undetermined and inconclusive verdicts of Section 6 were therefore forced by the data, not chosen by the method.* We read these counts as lower bounds, on the grounds that injected faults separate more cleanly than real ones; that is an argument, not a measurement, since it assumes stabilisation is monotone in separability.

8.4. Discriminative structure is joint-resolved

Holding the statistic fixed and varying only its resolution within the load-torque group, a mode-level scalar reaches 0.455, below the 0.504 majority rate; per signal family 0.527; joint-resolved over the load joints 0.719; joint-resolved over all six joints 0.860. FMEA prose cannot express the quantity that decides the group.

9. Conclusion

The residue a ranked candidate list leaves on this benchmark is structural, not statistical: the knowledge base's fault signature matrix has repeated columns, and 68.1% of anomalies tie at rank one as a result. Reporting a point answer under those conditions reports a tie-break rule. We measured how much: an arbitrary list order moves the reported top-1 across an interval several times wider than the margins these comparisons turn on, while the group assignment does not move at all.

What follows is a change of output form. Report the non-isolable group; ask which discarded dimension would separate it; issue the answer as a certificate carrying held-out evidence, controls and a cost class, inactive until authorised. One of four groups clears that bar on real data. The same analysis bounds the gain: cluster accuracy is 0.342, so resolving every tie perfectly would not carry end-to-end accuracy past it, and the errors are not confined to the groups that generate the ties.

The result we would most want carried forward is the protocol one. Under an ungrouped split the same pipeline certifies a collision repair that grouped validation rejects, and permuted-label controls do not catch it. Any diagnostic method that proposes to change what a plant measures should be held to evidence from fault instantiations it has never seen.

For a safety function built on a learned component, the property to establish first is not the accuracy of the candidate list but whether the list is decidable at all, and that, as the structural tradition holds, can be checked before any data is seen.

Code and data. Evidence records, certificates, the tie-break sweep and the producing scripts are archived at `https://doi.org/10.5281/zenodo.22089173`.

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